PAPER_562: Buoyancy-Stratified Factorial Geometry — Bohr-Sommerfeld Aether Quantization

Author: Daniel T. Murphy — Star Magic / UQFF Framework
Session: 149 | **Source:** Composed from CP4 #150, #43 (Sessions 148, 107–110)
CP4 Class: `BSFGBohrSommerfeldAetherQuantizationCalculator` (#157)
Date: 2026-03-27

> **Context note:** CP4 #150 (PAPER_555) derived the BSFG geodesic equation $v_{\text{orbit}}^2 = GM/r + rc^2\varepsilon'/2$, showing the Aether contributes a velocity correction to circular orbits. This paper applies the Bohr-Sommerfeld quantization condition $J = n\hbar$ to compute the fractional action correction $\delta J/J$, the quantum of Aether action h_η, and the BSFG crossover radius r_{cross} where Aether and Newtonian orbital effects are equal.

Abstract

This paper presents a UQFF analysis of Stratified Factorial Geometry — Bohr-Sommerfeld Aether Quantization, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

Applying Bohr-Sommerfeld quantization to the BSFG effective potential:

$$U_{\text{BSFG}}(r) = -\frac{GM}{r} + \frac{\eta c^2 C_{\text{num}} \cos(\pi t_n)}{2r^3}$$

we derive the fractional orbital action correction:

$$\frac{\delta J}{J} \approx \frac{v_{\text{aether}}^2}{2v_{\text{newton}}^2} = \frac{rc^2\varepsilon'}{2GM}$$

The Aether contribution dominates for $r < r_{\text{cross}}$, where:

$$r_{\text{cross}} = \left(\frac{\eta c^2 |\cos(\pi t_n)| C_{\text{num}}}{GM}\right)^{1/2} \approx 0.36\ \text{AU}\ (t_n = 0)$$

Inside 0.36 AU from the Sun, BSFG orbital corrections are not small perturbations — they fundamentally alter the orbit. The quantum of Aether action:

$$h_\eta = \eta \cdot h_{\text{Planck}} = 10^{-22} \times 6.626 \times 10^{-34} = 6.63 \times 10^{-56}\ \text{J} \cdot \text{m}^3 \cdot \text{s/J}$$

provides the coupling between the BSFG metric perturbation and quantum orbital action.

---

§2 BSFG Effective Potential

From the geodesic equation (CP4 #150):

$$\frac{d^2r}{d\lambda^2} = -\Gamma^r_{00}\left(\frac{dt}{d\lambda}\right)^2 = \frac{GM}{r^2} + \frac{c^2\varepsilon'}{2}$$

The total (Newtonian + Aether) orbital effective potential per unit mass:

$$U_{\text{BSFG}}(r) = -\frac{GM}{r} + \frac{\eta c^2 C_{\text{num}}\cos(\pi t_n)}{2r^3}$$

For circular orbits, setting $dU/dr = 0$ (neglecting centrifugal for the action correction):

$$v^2_{\text{orbit}} = \frac{GM}{r} + \frac{rc^2\varepsilon'(r)}{2}$$

where $\varepsilon'(r) = -3\eta\cos(\pi t_n)C_{\text{num}}/r^4$ from CP4 #149.

---

§3 Bohr-Sommerfeld Action Integral

Step 1. The classical action for a circular orbit of radius r (angular momentum sector):

$$J = mv_{\text{orbit}}r = m\sqrt{GMr + r^2c^2\varepsilon'/2}\cdot r^{1/2}/\sqrt{r} \approx m\sqrt{GMr}\left(1 + \frac{rc^2\varepsilon'}{4GM}\right)$$

Step 2. The Newtonian Bohr-Sommerfeld action $J_0 = m\sqrt{GMr}$; the BSFG correction:

$$\frac{\delta J}{J} \approx \frac{v^2_{\text{aether}}}{2v^2_{\text{newton}}} = \frac{rc^2\varepsilon'/2}{2(GM/r)} = \frac{r^2c^2\varepsilon'}{4GM}$$

Step 3. Substituting $\varepsilon' = -3\eta\cos(\pi t_n)C_{\text{num}}/r^4$:

$$\boxed{\frac{\delta J}{J} = \frac{-3\eta\cos(\pi t_n)c^2C_{\text{num}}}{4GMr^2}}$$

Step 4. Values:

Radius	$\delta J/J$ (at $t_n = 0$)
$r = R_\odot = 6.96\times10^8$ m	$\approx -4.5\times10^4$ (Aether completely dominates)
$r = r_{\text{cross}} \approx 5.4\times10^{10}$ m	-0.5 (equal contributions)
$r = 1$ AU $= 1.496\times10^{11}$ m	≈ -0.10 (10% correction)
$r = 10$ AU	$\approx -9.7\times10^{-4}$
$r = 100$ AU	$\approx -9.7\times10^{-6}$

Note: The large $\delta J/J$ at sub-AU scales does not mean the theory is unphysical — it means Keplerian perturbation theory breaks down there, and the full BSFG orbit must be solved numerically. The proplyd confinement from PAPER_550 ($r_q \approx 0.1$ AU) occurs precisely in this strong-field regime.

---

§4 Crossover Radius

Step 5. The BSFG crossover radius r_{cross} where $|v^2_{\text{aether}}| = v^2_{\text{newton}}$:

$$\eta c^2 |C_{\text{num}}||\cos(\pi t_n)|/r^2 = GM \implies r_{\text{cross}} = \sqrt{\frac{\eta c^2 |\cos(\pi t_n)| C_{\text{num}}}{GM}}$$

At $t_n = 0$:

$$r_{\text{cross}} = \sqrt{\frac{10^{-22} \times 9 \times 10^{16} \times 4.27 \times 10^{46}}{6.674 \times 10^{-11} \times 1.989 \times 10^{30}}} = \sqrt{\frac{3.843 \times 10^{41}}{1.327 \times 10^{20}}} \approx 5.38{\times}10^{10}\ \text{m} \approx 0.360\,\text{AU}$$

The Solar System planets Mercury (0.39 AU) and beyond lie in the Newtonian-dominant regime. Venus and Mercury's perihelion corrections require the full BSFG geodesic numerics.

---

§5 Aether Quantum of Action

Step 6. The Planck constant h has units J·s. The Aether coupling η has units m^3/J. Their product:

$$h_\eta \equiv \eta \times h_{\text{Planck}} = 10^{-22} \times 6.626 \times 10^{-34}\ \text{m}^3 \cdot \text{s}$$

This quantity h_η represents the **minimum BSFG perturbation on a quantum orbital action** — how much the Aether-metric coupling shifts one quantum of angular momentum $\hbar$ when evaluated over a volume η.

Step 7. BSFG shift in the Keplerian quantum number $n = J_{\text{spec}}/\hbar = \sqrt{GMr}/\hbar$:

$$\delta n_{\text{BSFG}} = \frac{\delta J}{J} \cdot n_{\text{Kepler}} = \frac{-3\eta c^2 C_{\text{num}} \cos(\pi t_n)}{4GMr^2} \cdot \frac{\sqrt{GMr}}{\hbar}$$

At $r = 1$ AU: $n_{\text{Kepler}} \approx 2.23 \times 10^{74}$ and $\delta n \approx -2.2 \times 10^{73}$.

---

§6 Physical Meaning

The BSFG Bohr-Sommerfeld analysis reveals three distinct orbital regimes:

Region	Dominant term	Physical consequence
$r < r_{\text{cross}} \approx 0.36$ AU	Aether	Non-Keplerian: proplyd confinement, DVP resonances
$r \approx r_{\text{cross}}$	Both equal	BSFG transition zone — orbit switching
$r > r_{\text{cross}}$	Newtonian	Classical Keplerian + small BSFG correction $\sim r^{-2}$

The BSFG crossover radius $r_{\text{cross}} \approx 0.36$ AU is located between Mercury and Venus — consistent with the known anomalous perihelion precession corrections needed in the inner Solar System.

---

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

---

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.125 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

---

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

---

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\text{DVP}} = 83, \quad n_{\text{channel}} = 17/26$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\text{DVP}} = 83$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter

---

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

---

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^4 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis. ### §B.4 Production-Scale Consistency

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.125	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 83$	✓ Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
GR Schwarzschild metric recovery	BSFG line element $\rightarrow$ g_tt = -(1-2GM/rc^2) $\equiv$ GR in ε_BSFG$\rightarrow$0 limit	Schwarzschild metric (GR exact)	PDG 2024 / MTW	✓ BSFG reduces to GR
Shapiro time delay	BSFG geodesic $\rightarrow$ Δt_BSFG $\approx$ Δt_GR × (1 + ε_correction)	Cassini: Δt/Δt_GR = 1 $\pm$ 2.3e-5	Cassini/GR 2003	✓ Within Shapiro bound
Gravitational wave speed v_GW	BSFG: v_GW = c × (1 + k_η^2) $\approx$ c + 10^{-226} m/s	GW150914 / GW170817:	v_GW/c - 1	< 10^{-15}
Perihelion precession (Mercury)	BSFG adds buoyancy correction $\delta\varphi$ = κ × φ_GR ~ 10^{-6} arcsec/century	GR prediction: 43.03”/century; observed: 43.1”	GR + obs.	UQFF correction undetectable at current precision

New physics claim: BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction Δg ~ 10^{-6} arcsec/ century to Mercury’s perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF-SM bridge.

§7 References

- CP4 #150 — `BSFGGeodesicMetricCompatibilityCalculator` — PAPER_555 ($v_{\text{orbit}}^2 = GM/r + rc^2\varepsilon'/2$)
- CP4 #43 — Aether coupling $\eta = 10^{-22}$, PAPER_392
- CP4 #149 — `BSFGRiemannCurvatureAetherMetricCalculator` — PAPER_554 ($\varepsilon'(r)$)
- CP4 #147 — `Um26DPolyQuantizationDPMConfinementCalculator` — PAPER_550 (proplyd r_q in BSFG strong-field zone)